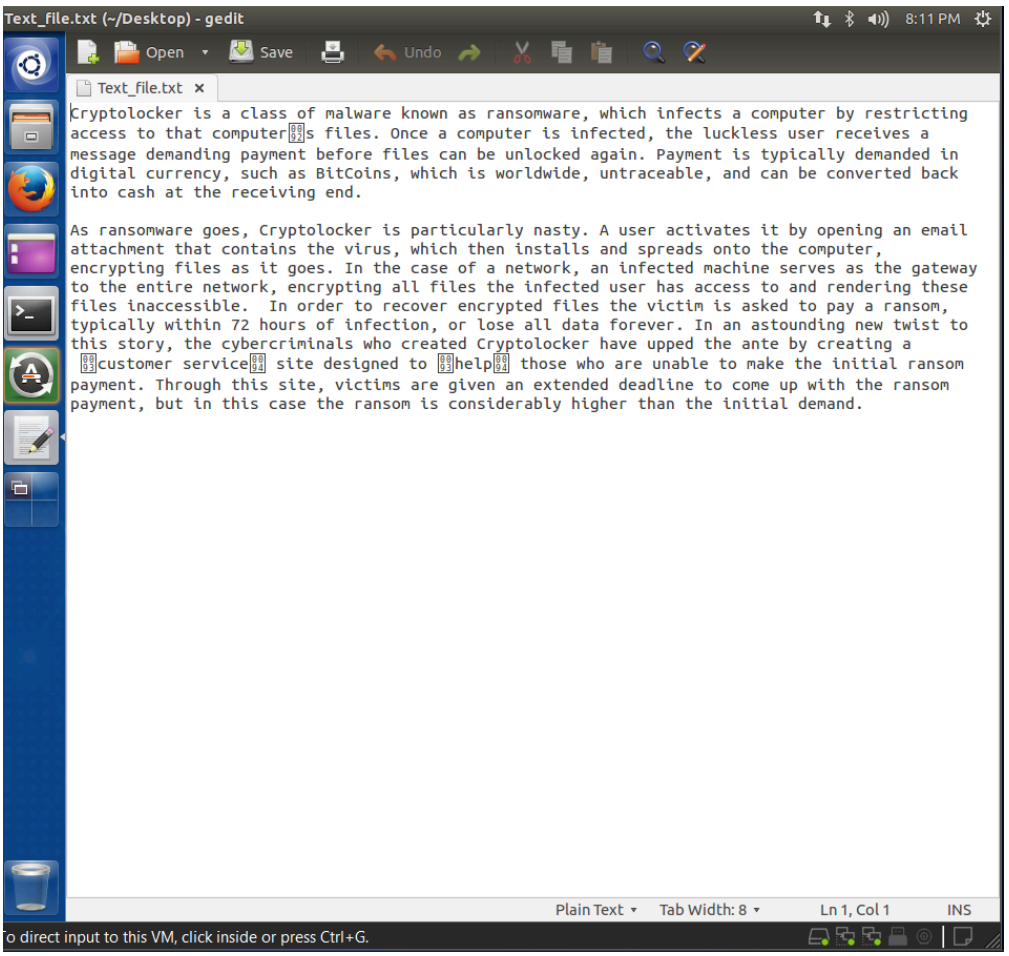
CYBERSECURITY LAB ASSIGNMENT 03

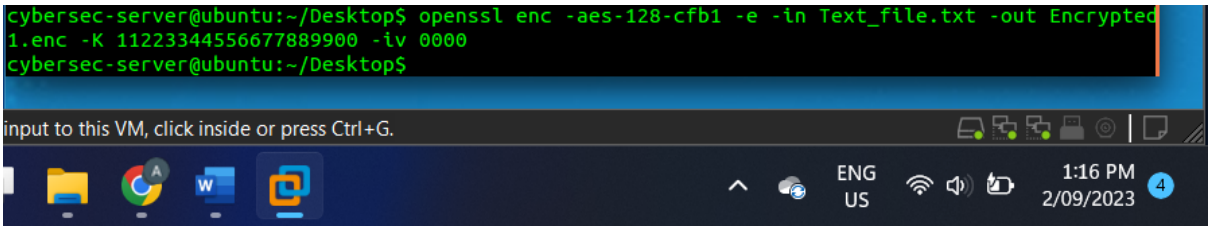
Task-1: Basic encryption and decryption using OpenSSL.

▪ 2 times encryption on Text_file using different ciphers and compare the output, i.e., encrypted texts



Text file before encryption

Image of the encrypted file encrypted with cipher text aes 128 cfb1



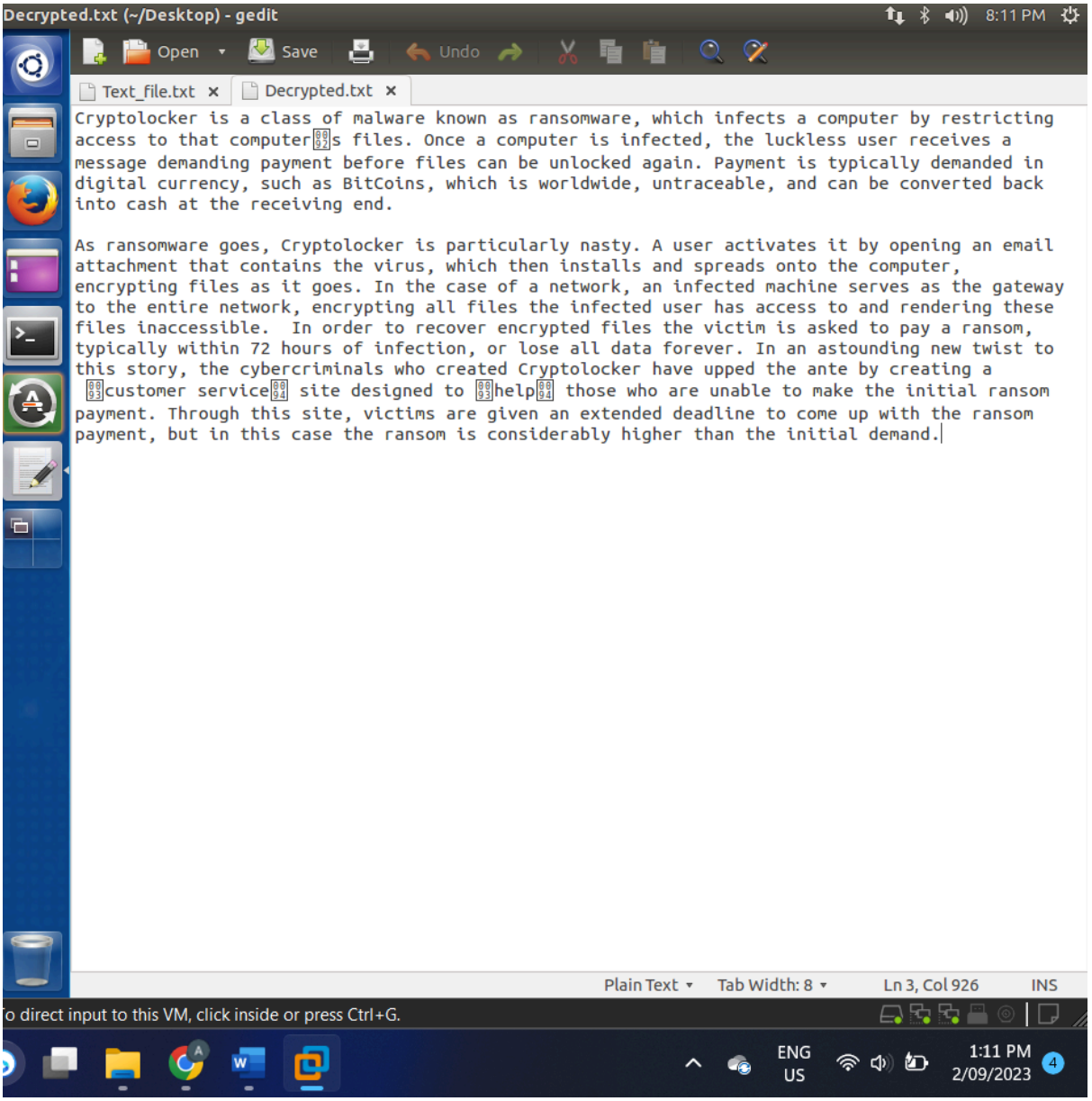


Image of the decrypted file

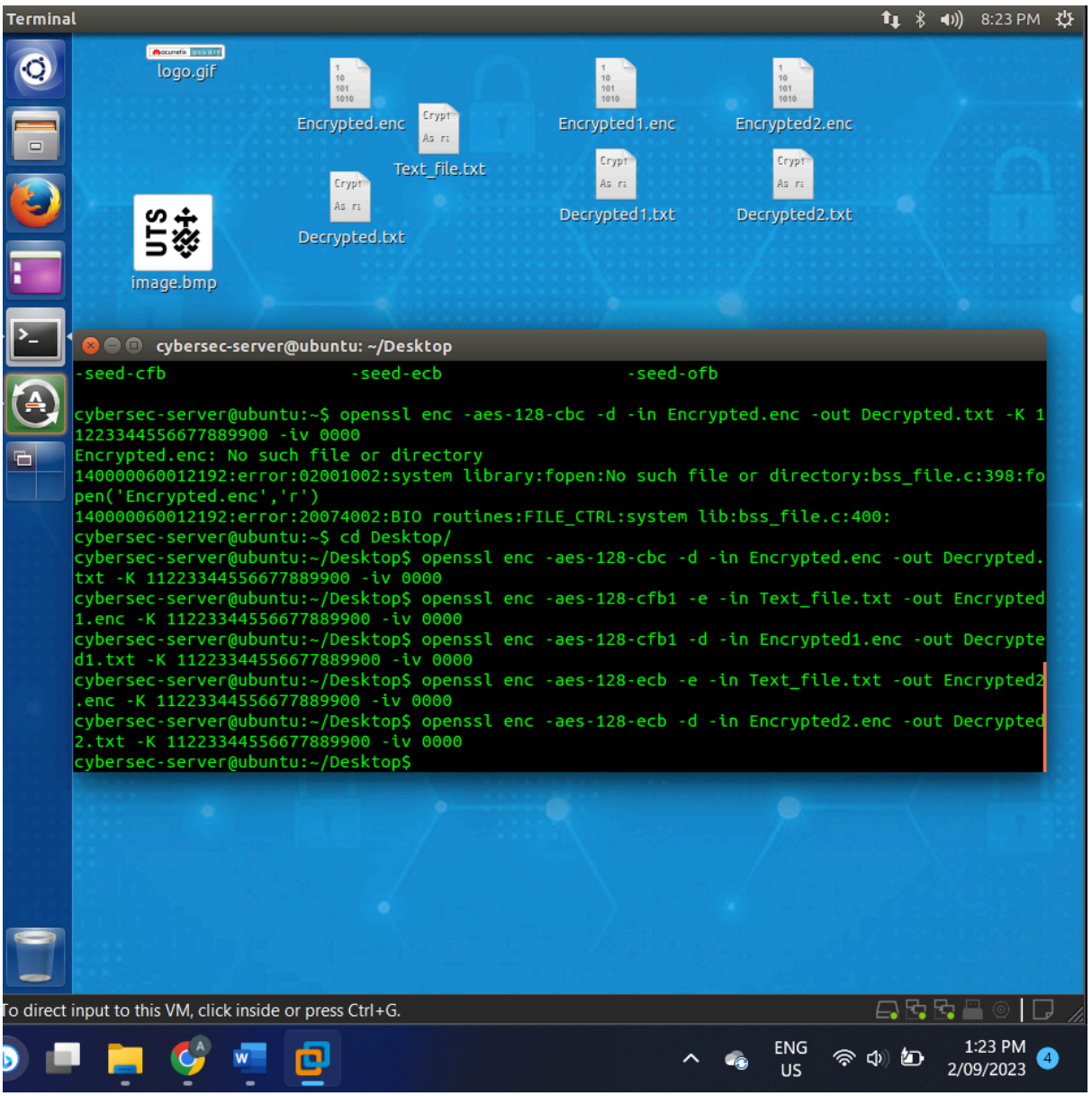
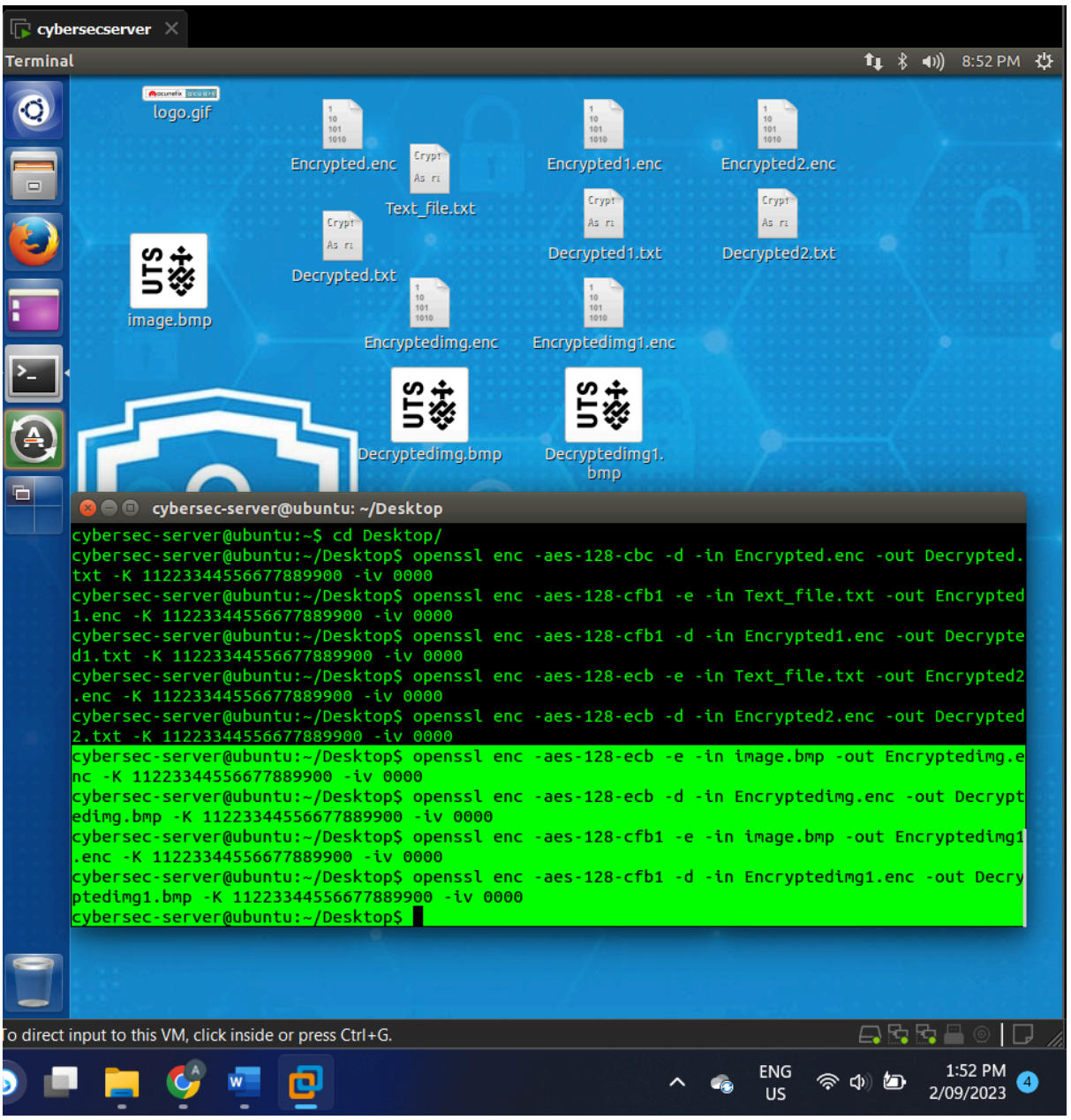


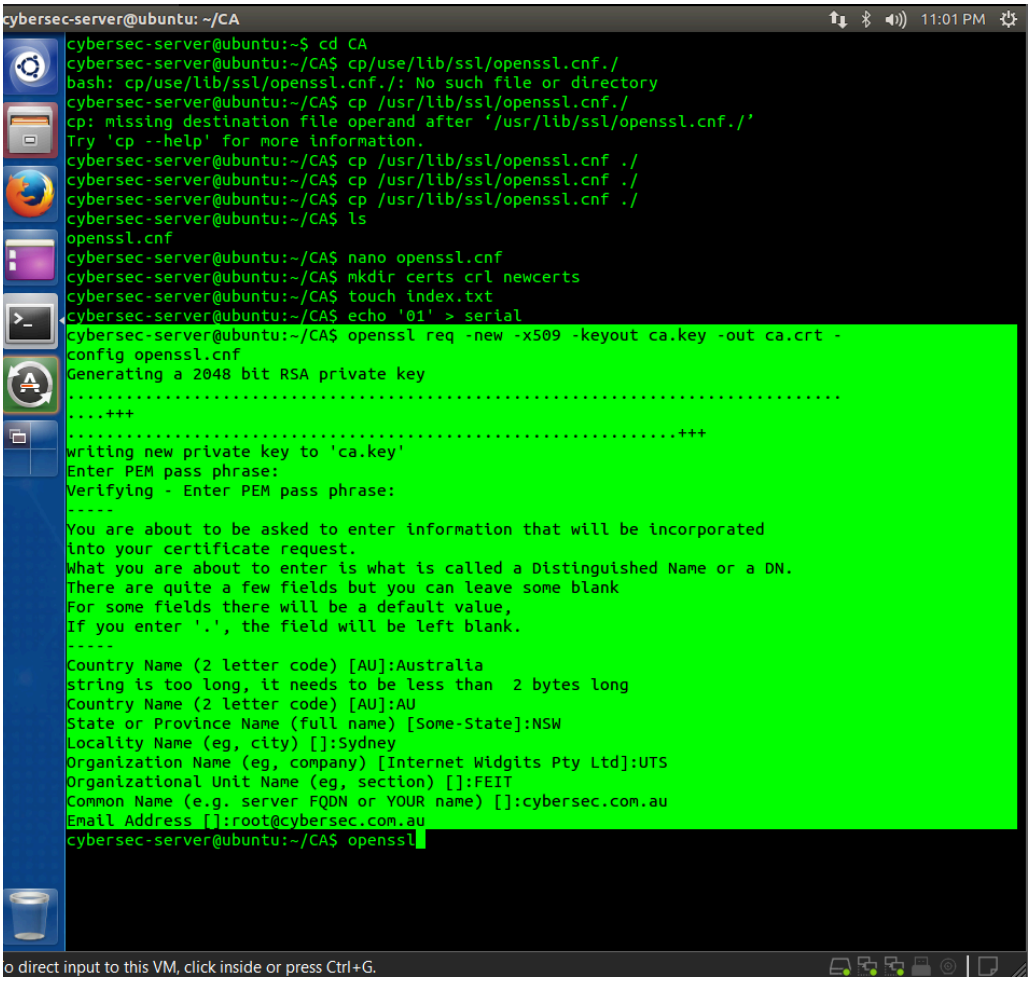
Image of encrypted2.enc file and the corresponding decrypted file. Encrypted with cipher text aes 128 ecb.

▪ 2 times encrypt the image file using different ciphers and compare the output, i.e., encrypted texts

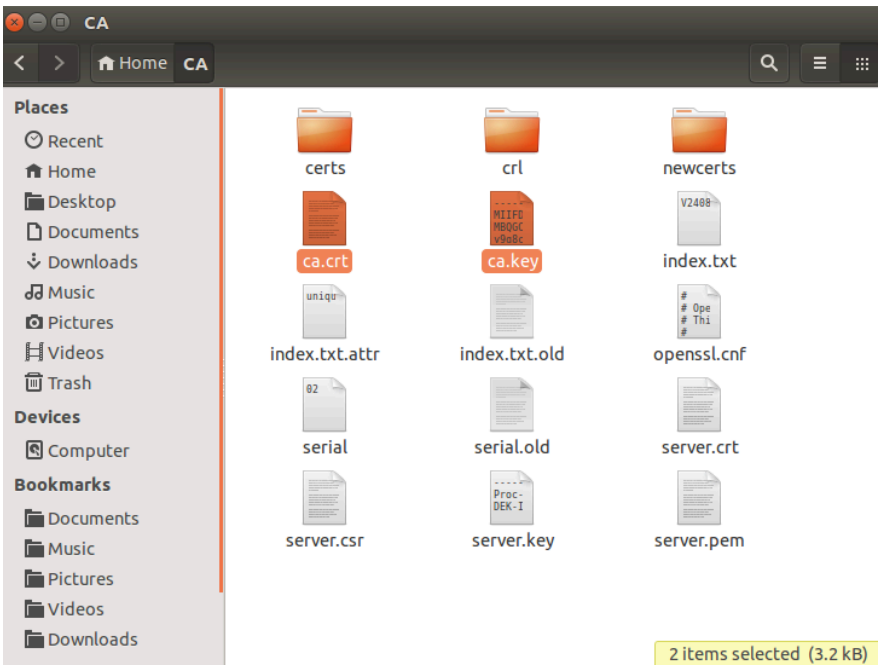


Encryption and decryption of image file using aes-128-ecb and aes-128 cfb1 ciphertext.

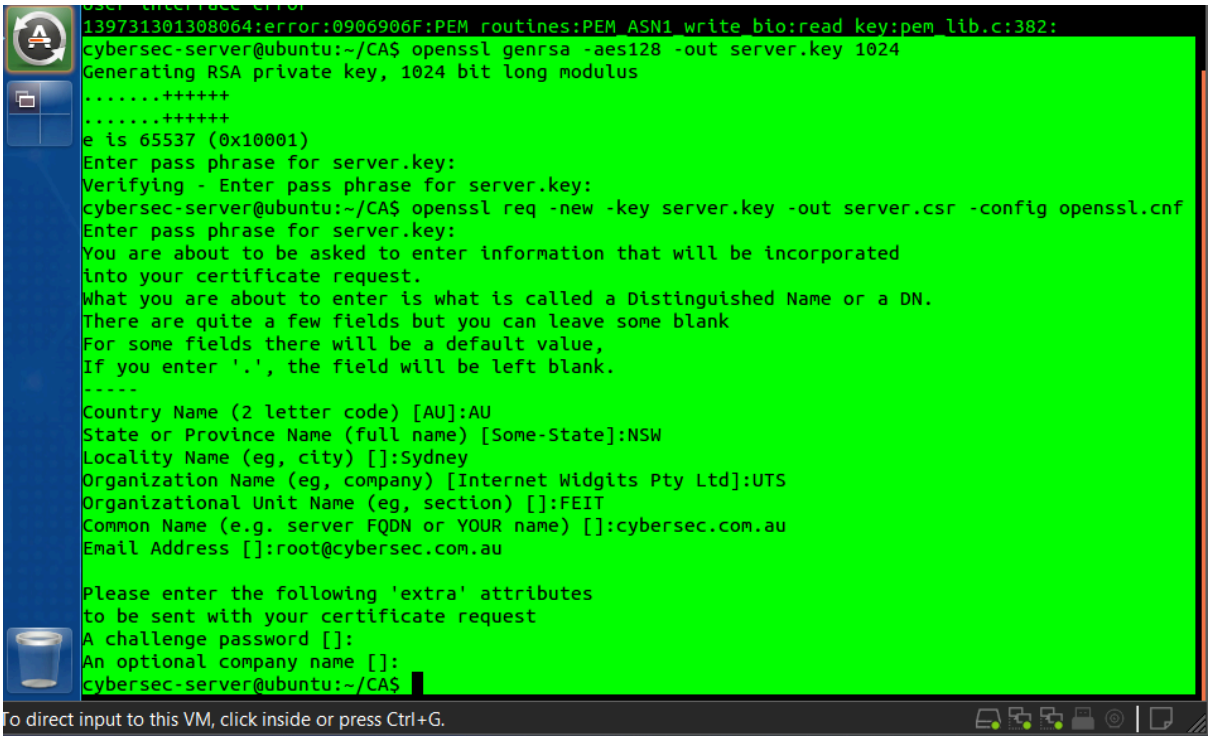
Task-2: Become a CA and generate a self-signed certificate



```
cybersec-server@ubuntu: ~/CA                                    11:01 PM
cybersec-server@ubuntu:~$ cd CA
cybersec-server@ubuntu:~/CA$ cp/use/lib/ssl/openssl.cnf./
bash: cp/use/lib/ssl/openssl.cnf./: No such file or directory
cybersec-server@ubuntu:~/CA$ cp /usr/lib/ssl/openssl.cnf./
cp: missing destination file operand after '/usr/lib/ssl/openssl.cnf./'
Try 'cp --help' for more information.
cybersec-server@ubuntu:~/CA$ cp /usr/lib/ssl/openssl.cnf ./
cybersec-server@ubuntu:~/CA$ cp /usr/lib/ssl/openssl.cnf ./
cybersec-server@ubuntu:~/CA$ cp /usr/lib/ssl/openssl.cnf ./
cybersec-server@ubuntu:~/CA$ ls
openssl.cnf
cybersec-server@ubuntu:~/CA$ nano openssl.cnf
cybersec-server@ubuntu:~/CA$ mkdir certs crl newcerts
cybersec-server@ubuntu:~/CA$ touch index.txt
cybersec-server@ubuntu:~/CA$ echo '01' > serial
cybersec-server@ubuntu:~/CA$ openssl req -new -x509 -keyout ca.key -out ca.crt -
config openssl.cnf
Generating a 2048 bit RSA private key
.......................................................................
....+++
.............................................................+++
writing new private key to 'ca.key'
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:Australia
string is too long, it needs to be less than  2 bytes long
Country Name (2 letter code) [AU]:AU
State or Province Name (full name) [Some-State]:NSW
Locality Name (eg, city) []:Sydney
Organization Name (eg, company) [Internet Widgits Pty Ltd]:UTS
Organizational Unit Name (eg, section) []:FEIT
Common Name (e.g. server FQDN or YOUR name) []:cybersec.com.au
Email Address []:root@cybersec.com.au
cybersec-server@ubuntu:~/CA$ openssl
o direct input to this VM, click inside or press Ctrl+G.
```



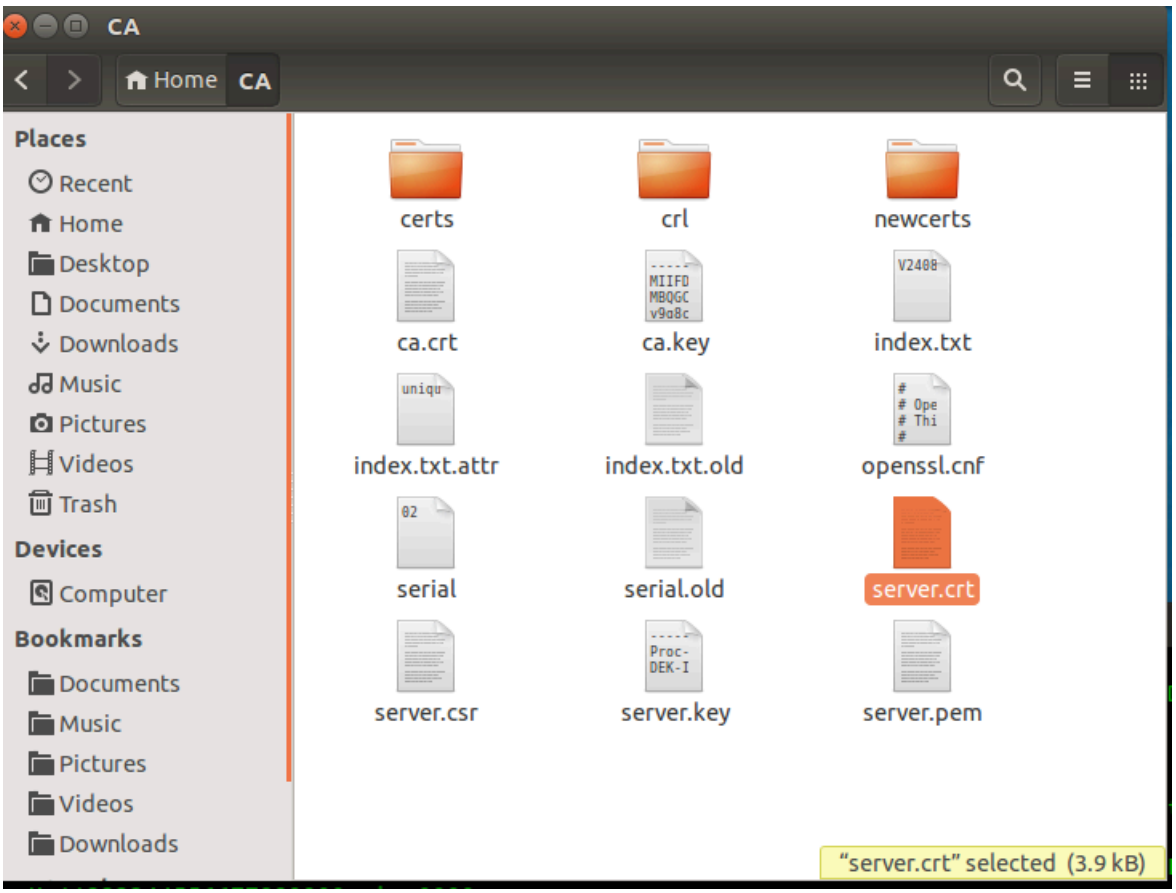
Task-3: Create a Certificate for cybersec.com.au as a Root CA



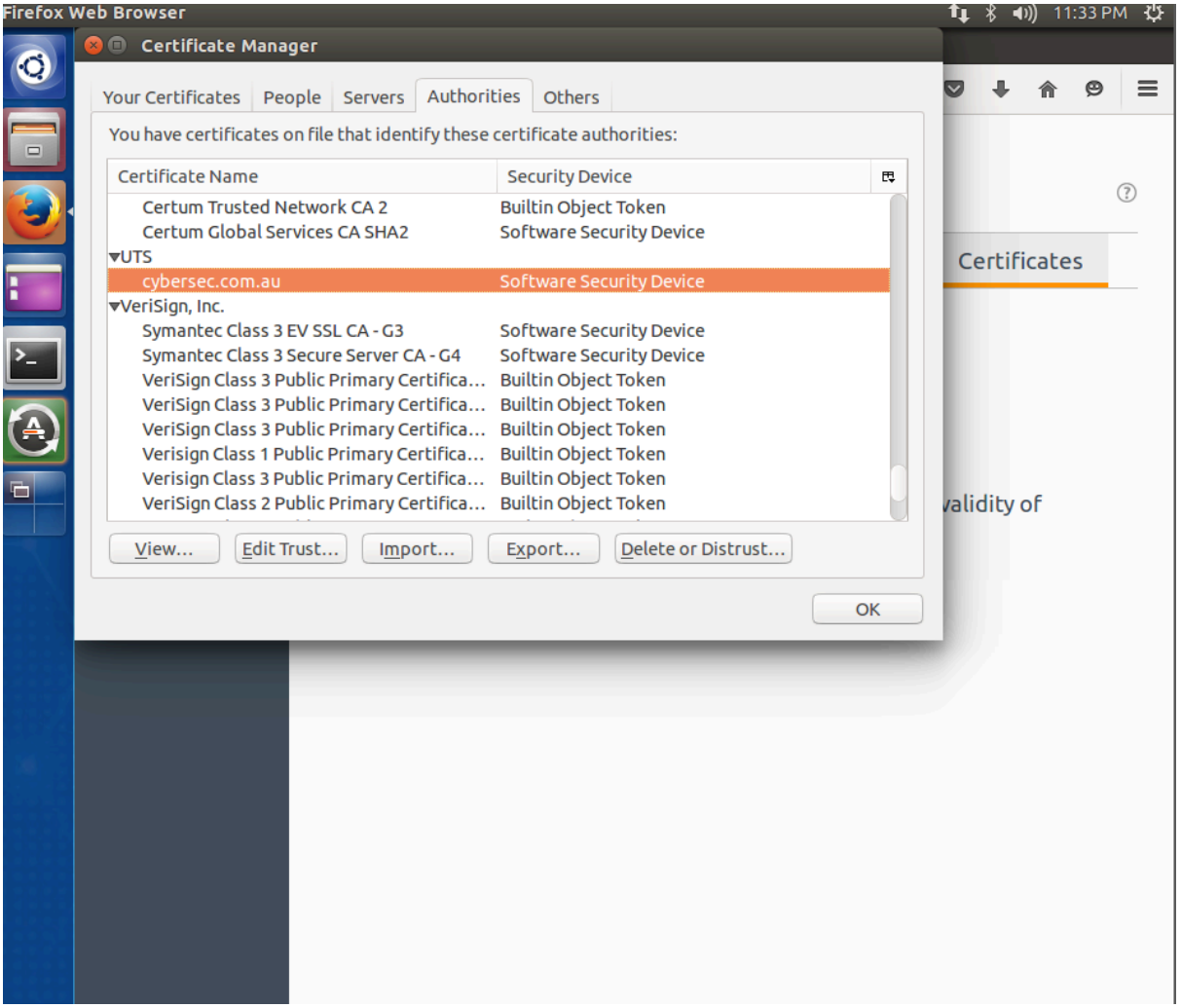
```
139731301308064:error:0906906F:PEM routines:PEM_ASN1_write_bio:read key:pem_lib.c:382:
cybersec-server@ubuntu:~/CA$ openssl genrsa -aes128 -out server.key 1024
Generating RSA private key, 1024 bit long modulus
.......++++++
.......++++++
e is 65537 (0x10001)
Enter pass phrase for server.key:
Verifying - Enter pass phrase for server.key:
cybersec-server@ubuntu:~/CA$ openssl req -new -key server.key -out server.csr -config openssl.cnf
Enter pass phrase for server.key:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:AU
State or Province Name (full name) [Some-State]:NSW
Locality Name (eg, city) []:Sydney
Organization Name (eg, company) [Internet Widgits Pty Ltd]:UTS
Organizational Unit Name (eg, section) []:FEIT
Common Name (e.g. server FQDN or YOUR name) []:cybersec.com.au
Email Address []:root@cybersec.com.au

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:
cybersec-server@ubuntu:~/CA$
```

To direct input to this VM, click inside or press Ctrl+G.



Task-4: Use PKI for websites


Firefox Web Browser
11:33 PM
Certificate Manager
Your Certificates
People
Servers
Authorities
Others
You have certificates on file that identify these certificate authorities:
Certificate Name
Security Device
Certum Trusted Network CA 2
Builtin Object Token
Certum Global Services CA SHA2
Software Security Device
UTS
cybersec.com.au
Software Security Device
VeriSign, Inc.
Symantec Class 3 EV SSL CA - G3
Software Security Device
Symantec Class 3 Secure Server CA - G4
Software Security Device
VeriSign Class 3 Public Primary Certifica...
Builtin Object Token
VeriSign Class 3 Public Primary Certifica...
Builtin Object Token
VeriSign Class 3 Public Primary Certifica...
Builtin Object Token
Verisign Class 1 Public Primary Certifica...
Builtin Object Token
Verisign Class 3 Public Primary Certifica...
Builtin Object Token
VeriSign Class 2 Public Primary Certifica...
Builtin Object Token
View...
Edit Trust...
Import...
Export...
Delete or Distrust...
OK
Certificates
validity of

Mozilla Firefox 11:33 PM

https://cyb...om.au:4433/ × Preferences ×

https://cybersec.com.au:4433 Search

```
s_server -cert server.pem -www
Secure Renegotiation IS supported
Ciphers supported in s_server binary
TLSv1/SSLv3:ECDHE-RSA-AES256-GCM-SHA384TLSv1/SSLv3:ECDHE-ECDSA-AES256-GCM-SHA384
TLSv1/SSLv3:ECDHE-RSA-AES256-SHA384  TLSv1/SSLv3:ECDHE-ECDSA-AES256-SHA384
TLSv1/SSLv3:ECDHE-RSA-AES256-SHA     TLSv1/SSLv3:ECDHE-ECDSA-AES256-SHA
TLSv1/SSLv3:SRP-DSS-AES-256-CBC-SHA  TLSv1/SSLv3:SRP-RSA-AES-256-CBC-SHA
TLSv1/SSLv3:SRP-AES-256-CBC-SHA      TLSv1/SSLv3:DHE-DSS-AES256-GCM-SHA384
TLSv1/SSLv3:DHE-RSA-AES256-GCM-SHA384TLSv1/SSLv3:DHE-RSA-AES256-SHA256
TLSv1/SSLv3:DHE-DSS-AES256-SHA256    TLSv1/SSLv3:DHE-RSA-AES256-SHA
TLSv1/SSLv3:DHE-DSS-AES256-SHA       TLSv1/SSLv3:DHE-RSA-CAMELLIA256-SHA
TLSv1/SSLv3:DHE-DSS-CAMELLIA256-SHA  TLSv1/SSLv3:ECDH-RSA-AES256-GCM-SHA384
TLSv1/SSLv3:ECDH-ECDSA-AES256-GCM-SHA384TLSv1/SSLv3:ECDH-RSA-AES256-SHA384
TLSv1/SSLv3:ECDH-ECDSA-AES256-SHA384 TLSv1/SSLv3:ECDH-RSA-AES256-SHA
TLSv1/SSLv3:ECDH-ECDSA-AES256-SHA    TLSv1/SSLv3:AES256-GCM-SHA384
TLSv1/SSLv3:AES256-SHA256            TLSv1/SSLv3:AES256-SHA
TLSv1/SSLv3:CAMELLIA256-SHA          TLSv1/SSLv3:PSK-AES256-CBC-SHA
TLSv1/SSLv3:ECDHE-RSA-DES-CBC3-SHA   TLSv1/SSLv3:ECDHE-ECDSA-DES-CBC3-SHA
TLSv1/SSLv3:SRP-DSS-3DES-EDE-CBC-SHA TLSv1/SSLv3:SRP-RSA-3DES-EDE-CBC-SHA
TLSv1/SSLv3:SRP-3DES-EDE-CBC-SHA     TLSv1/SSLv3:EDH-RSA-DES-CBC3-SHA
TLSv1/SSLv3:EDH-DSS-DES-CBC3-SHA     TLSv1/SSLv3:ECDH-RSA-DES-CBC3-SHA
TLSv1/SSLv3:ECDH-ECDSA-DES-CBC3-SHA  TLSv1/SSLv3:DES-CBC3-SHA
TLSv1/SSLv3:PSK-3DES-EDE-CBC-SHA     TLSv1/SSLv3:ECDHE-RSA-AES128-GCM-SHA256
TLSv1/SSLv3:ECDHE-ECDSA-AES128-GCM-SHA256TLSv1/SSLv3:ECDHE-RSA-AES128-SHA256
TLSv1/SSLv3:ECDHE-ECDSA-AES128-SHA256TLSv1/SSLv3:ECDHE-RSA-AES128-SHA
TLSv1/SSLv3:ECDHE-ECDSA-AES128-SHA   TLSv1/SSLv3:SRP-DSS-AES-128-CBC-SHA
TLSv1/SSLv3:SRP-RSA-AES-128-CBC-SHA  TLSv1/SSLv3:SRP-AES-128-CBC-SHA
TLSv1/SSLv3:DHE-DSS-AES128-GCM-SHA256TLSv1/SSLv3:DHE-RSA-AES128-GCM-SHA256
TLSv1/SSLv3:DHE-RSA-AES128-SHA256    TLSv1/SSLv3:DHE-DSS-AES128-SHA256
TLSv1/SSLv3:DHE-RSA-AES128-SHA       TLSv1/SSLv3:DHE-DSS-AES128-SHA
TLSv1/SSLv3:DHE-RSA-SEED-SHA         TLSv1/SSLv3:DHE-DSS-SEED-SHA
TLSv1/SSLv3:DHE-RSA-CAMELLIA128-SHA  TLSv1/SSLv3:DHE-DSS-CAMELLIA128-SHA
TLSv1/SSLv3:ECDH-RSA-AES128-GCM-SHA256TLSv1/SSLv3:ECDH-ECDSA-AES128-GCM-SHA256
TLSv1/SSLv3:ECDH-RSA-AES128-SHA256   TLSv1/SSLv3:ECDH-ECDSA-AES128-SHA256
TLSv1/SSLv3:ECDH-RSA-AES128-SHA      TLSv1/SSLv3:ECDH-ECDSA-AES128-SHA
TLSv1/SSLv3:AES128-GCM-SHA256        TLSv1/SSLv3:AES128-SHA256
TLSv1/SSLv3:AES128-SHA               TLSv1/SSLv3:SEED-SHA
TLSv1/SSLv3:CAMELLIA128-SHA          TLSv1/SSLv3:PSK-AES128-CBC-SHA
TLSv1/SSLv3:ECDHE-RSA-RC4-SHA        TLSv1/SSLv3:ECDHE-ECDSA-RC4-SHA
TLSv1/SSLv3:ECDH-RSA-RC4-SHA         TLSv1/SSLv3:ECDH-ECDSA-RC4-SHA
TLSv1/SSLv3:RC4-SHA                  TLSv1/SSLv3:RC4-MD5
TLSv1/SSLv3:PSK-RC4-SHA              TLSv1/SSLv3:EDH-RSA-DES-CBC-SHA
TLSv1/SSLv3:EDH-DSS-DES-CBC-SHA      TLSv1/SSLv3:DES-CBC-SHA
---
Ciphers common between both SSL end points:
ECDHE-ECDSA-AES128-GCM-SHA256 ECDHE-RSA-AES128-GCM-SHA256 ECDHE-ECDSA-AES256-SHA
ECDHE-ECDSA-AES128-SHA     ECDHE-RSA-AES128-SHA       ECDHE-RSA-AES256-SHA
DHE-RSA-AES128-SHA         DHE-RSA-AES256-SHA         AES128-SHA
AES256-SHA                 DES-CBC3-SHA
---
New, TLSv1/SSLv3, Cipher is ECDHE-RSA-AES128-GCM-SHA256
```

o direct input to this VM, click inside or press Ctrl+G.